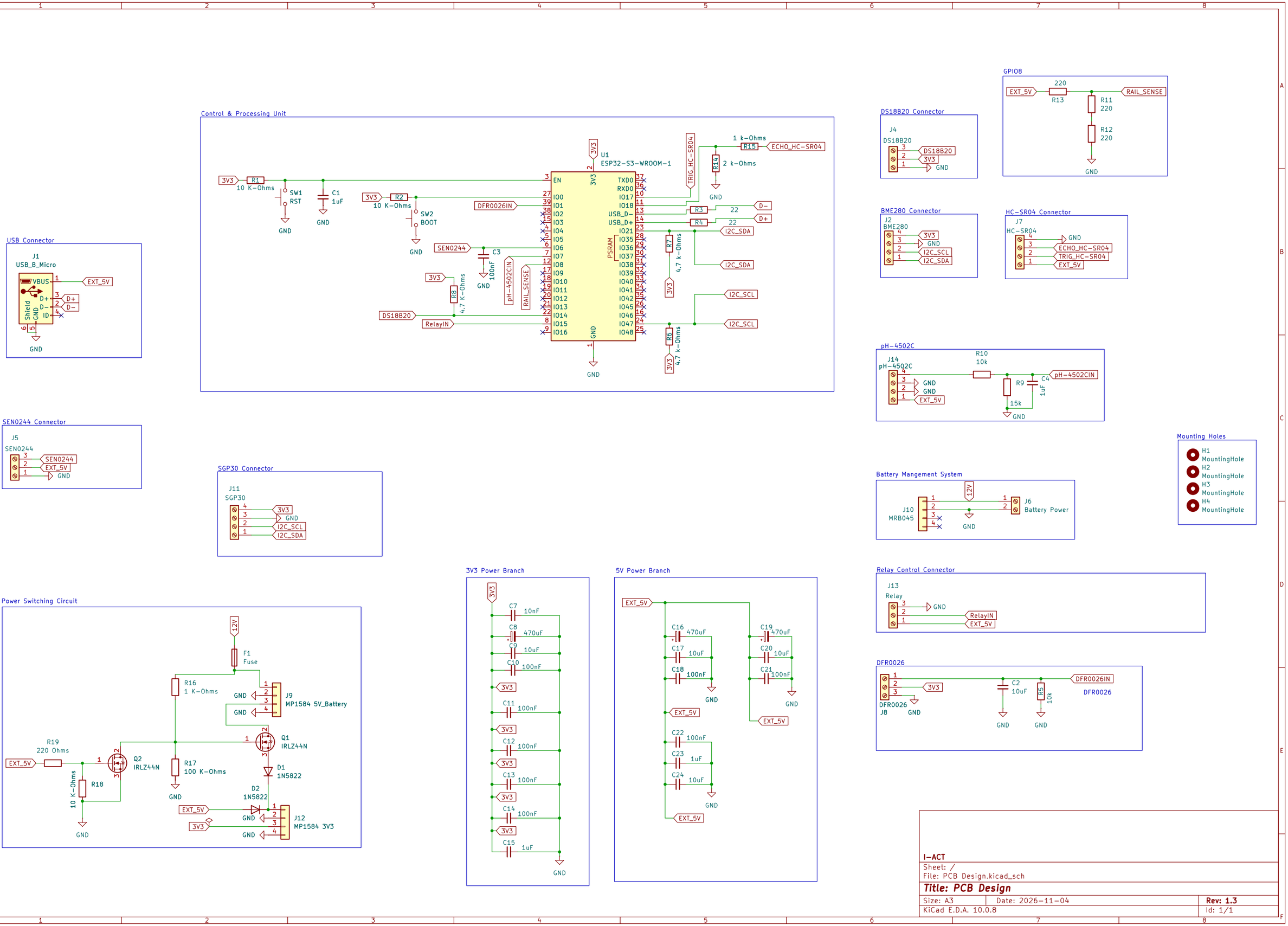

Control & Processing Unit
U1
ESP32-S3-WROOM-1
3V3
R1
10 K-Ohms
SW1
RST
C1
1uF
GND
R2
SW2
BOOT
DFR0026IN
SEN0244
C3
100nF
pH-4502CIN
RAIL_SENSE
R8
4.7 K-Ohms
DS18B20
RelayIN
TRIG_HC-SR04
R15
1 k-Ohms
ECHO_HC-SR04
R14
2 k-Ohms
R3
22
D-
R4
22
D+
I2C_SDA
I2C_SDA
R7
4.7 k-Ohms
I2C_SCL
I2C_SCL
R6
4.7 k-Ohms
EN
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
PSRAM
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
USB Connector
J1
USB_B_Micro
VBUS
D+
D-
ID
Shield
GND
EXT_5V
DS18B20 Connector
J4
DS18B20
GPIO8
220
R13
EXT_5V
R11
220
RAIL_SENSE
R12
220
BME280 Connector
J2
BME280
HC-SR04 Connector
J7
HC-SR04
pH-4502C
J14
R10
10k
R9
15k
C4
1uF
SEN0244 Connector
J5
SEN0244
SGP30 Connector
J11
SGP30
I2C_SCL
I2C_SDA
Mounting Holes
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Battery Mangement System
J10
MRB045
12V
J6
Battery Power
Power Switching Circuit
F1
Fuse
R16
1 K-Ohms
J9
MP1584 5V_Battery
R19
220 Ohms
Q1
IRLZ44N
Q2
IRLZ44N
R17
100 K-Ohms
R18
10 K-Ohms
D1
1N5822
D2
1N5822
J12
MP1584 3V3
3V3 Power Branch
C7
10nF
C8
470uF
C9
10uF
C10
100nF
C11
100nF
C12
100nF
C13
100nF
C14
100nF
C15
1uF
5V Power Branch
C16
470uF
C17
10uF
C18
100nF
C19
470uF
C20
10uF
C21
100nF
C22
100nF
C23
1uF
C24
10uF
Relay Control Connector
J13
Relay
RelayIN
EXT_5V
DFR0026
J8
C2
10uF
R5
10k
DFR0026IN
I-ACT
Sheet: /
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Title: PCB Design
Size: A3
Date: 2026-11-04
Rev: 1.3
KiCad E.D.A. 10.0.8
Id: 1/1